AP BIOLOGY TOPIC 8.7

Disruptions to Ecosystems: Invasive Species, Eutrophication & Biomagnification

UNIT 8: ECOLOGY (10–15% EXAM WEIGHT)

EVO-1.O | SYI-2.A | SYI-2.B | SYI-2.C | INVASIVE SPECIES

Credits: mitali120501@gmail.com

§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 8.7 (DISRUPTIONS TO ECOSYSTEMS: INVASIVE SPECIES, EUTROPHICATION & BIOMAGNIFICATION)

CED Core Vocabulary

Mutations Are Random (Not Directed by Need!) — The environment does NOT create beneficial mutations on demand.

Invasive Species — Non-native (exotic/introduced) species intentionally or accidentally transported into a new ecosystem where they spread rapidly and disrupt native communities.

Kudzu Vine (SE US) — Introduced from Japan to prevent soil erosion → grows up to 1 ft/day, smothering native forest trees and blocking sunlight.

Cane Toads (Australia) — Introduced to eat cane beetles → secrete fatal cardiotoxic bufotoxin that kills native marsupial quolls, monitor lizards, and snakes that try to eat them.

Heterozygote Advantage Maintains Alleles — When the heterozygous genotype has higher fitness than either homozygote (e.g., Sickle-cell trait HbA/HbS providing malaria resistance in endemic regions),...

2. Exploit Unused Niches or Outcompete Natives — Superior resource extraction + naive native prey that lack co-evolved anti-predator defenses.

Zebra Mussels (Great Lakes) — Arrived in ship ballast water → filter out phytoplankton, starving native mussels and clogging water intake pipes.

El Niño–Southern Oscillation (ENSO) — Warm Pacific surface water suppresses cold, nutrient-rich coastal upwelling along South America → phytoplankton crash → anchovy fish & seabird/marine mammal...

§2. 1. RANDOM MUTATIONS, ADAPTATIONS & ENVIRONMENTAL SHIFTS

CED EVO-1.O

- **An Adaptation Is a Genetic Variation Favored by Selection:** Increases survival or reproductive fitness in a particular environment.
- **Mutations Are Random (Not Directed by Need!):**
  - The environment **does NOT create** beneficial mutations on demand.
  - Random mutations generate genetic variation *prior* to environmental change; when an ecosystem is disrupted, **natural selection acts on pre-existing variation**!
- **Heterozygote Advantage Maintains Alleles:** When the heterozygous genotype has higher fitness than either homozygote (e.g., **Sickle-cell trait** **HbA/HbS** providing malaria resistance in endemic regions), harmful recessive alleles remain preserved in the gene pool.

[COMMON AP EXAM FRQ TRAP] **Never say organisms mutate "because they needed to adapt" to pollution or climate change!** Individuals with pre-existing tolerance alleles survive the disruption and reproduce; populations lacking those alleles decline or go extinct.

§3. 2. INVASIVE SPECIES: MECHANISMS & ECOLOGICAL IMPACTS

Unchecked Exponential Growth

- **Invasive Species:** Non-native (exotic/introduced) species intentionally or accidentally transported into a new ecosystem where they spread rapidly and disrupt native communities.
- **Why Invasive Species Succeed So Dramatically (3 Key Reasons):**
  - **1. Lack of Natural Predators, Parasites, or Pathogens:** Escaped top-down population control from their native range ("enemy release").
  - **2. Exploit Unused Niches or Outcompete Natives:** Superior resource extraction + naive native prey that lack co-evolved anti-predator defenses.
  - **3. High Reproductive Rate (*r*-Selected Traits):** Rapid maturation, huge brood sizes, and generalist diet/habitat tolerance → initial **exponential (J-curve) growth**!

[EXAMPLE & BIOLOGICAL APPLICATION] **High-Yield AP Invasive Species Case Studies:**

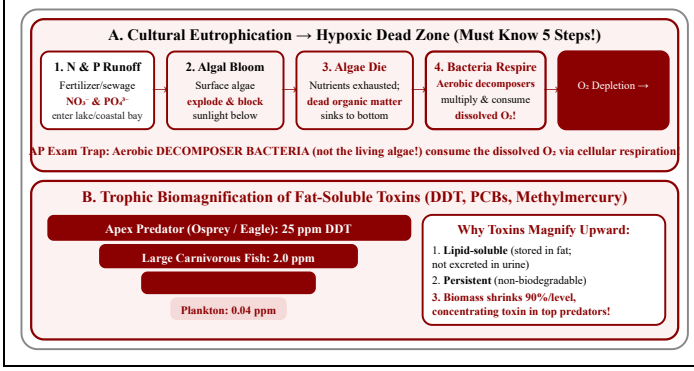
- **Kudzu Vine (SE US):** Introduced from Japan to prevent soil erosion → grows up to 1 ft/day, smothering native forest trees and blocking sunlight.
- **Zebra Mussels (Great Lakes):** Arrived in ship ballast water → filter out phytoplankton, starving native mussels and clogging water intake pipes.
- **Cane Toads (Australia):** Introduced to eat cane beetles → secrete fatal cardiotoxic bufotoxin that kills native marsupial quolls, monitor lizards, and snakes that try to eat them.
- **Brown Tree Snake (Guam) & Burmese Python (Everglades):** Decimated native island forest birds and Everglades marsh mammals.

§5. 4. HUMAN DISRUPTIONS: EUTROPHICATION & BIOMAGNIFICATION

High-Yield Step Pathways

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

4. Human Disruptions ➔ **Cultural Eutrophication** ➔ **Bioaccumulation vs. Biomagnification**



- **Cultural Eutrophication:** Agricultural fertilizer (**N, P**) and sewage runoff trigger explosive algal blooms; when algae die, **aerobic decomposer bacteria** consume dissolved oxygen (**O_2**) via cellular respiration, creating hypoxic **"dead zones"** (e.g., Gulf of Mexico) that suffocate fish and shellfish.
- **Bioaccumulation vs. Biomagnification:**
  - **Bioaccumulation:** Gradual buildup of a persistent toxin **within a single organism's tissues** over its lifetime.
  - **Biomagnification:** Increasing concentration of persistent, fat-soluble toxins (**DDT, PCBs, methylmercury**) in tissues at **successively higher trophic levels** in a food chain (e.g., DDT caused fragile eggshell thinning in bald eagles and peregrine falcons, documented in Rachel Carson's *Silent Spring*).

§4. 3. GEOLOGICAL, METEOROLOGICAL & CLIMATE DISRUPTIONS

Abiotic & Global Change

- **Geological & Meteorological Events Alter Habitats & Distributions:**
  - **El Niño–Southern Oscillation (ENSO):** Warm Pacific surface water suppresses cold, nutrient-rich coastal **upwelling** along South America → phytoplankton crash → anchovy fish & seabird/marine mammal populations plummet.
  - **Continental Drift & Meteor Impacts:** Shift climate zones, isolate clades, or trigger mass extinctions.
- **Anthropogenic Climate Change, Ocean Acidification & Habitat Loss:**
  - **Habitat Fragmentation:** Roads, logging, and agriculture split large habitats into small, isolated patches — reducing population size (**N↓**), blocking **gene flow**, increasing **genetic drift/inbreeding**, and creating harmful *edge effects* (solved by **wildlife corridors**!).
  - **Ocean Acidification:** Excess atmospheric **CO_2** dissolves in seawater to form carbonic acid (**$CO_2 + H_2O \leftrightarrow H_2CO_3 \leftrightarrow H^+ + HCO_3^-$**), lowering ocean pH and depleting carbonate ions (**CO_3^{2-}**) needed by corals and mollusks to build **$CaCO_3$** shells/skeletons; warm water also triggers **coral bleaching** (expulsion of mutualistic *zooxanthellae* algae).

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 8.7 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 8.7: Disruptions to Ecosystems: Invasive Species, Eutrophication & Biomagnification ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 10, 11 & 12 (Energy Dynamics, Whole-Plant Transpiration & Pillbug Animal Behavior Choice Chambers): Choice chambers testing pillbug (Armadillidium vulgare) chemotaxis/hydrotaxis/kinesis (count pillbugs on wet vs. dry filter paper every 30 s for 10 min; evaluate using χ^2 against 50:50 null distribution). Potometer/gravimetric transpiration across light, wind, and humidity treatments normalized per cm² leaf surface area. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Ecological Equations: Population Growth Rate: dN/dt = B − D. Exponential Growth: dN/dt = r_max·N (J-curve). Logistic Growth: dN/dt = r_max·N·[(K − N)/K] (S-curve; maximum instantaneous growth rate dN/dt occurs at N = K/2). Simpson's Diversity Index: D_I = 1 − Σ(n / N)² (ranges from 0 to 1; higher value = higher diversity). Net Primary Productivity: NPP = GPP − R. Trophic 10% Rule: ~10% of energy transfers to the next trophic level (~90% lost as metabolic heat/respiration). **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Ecological Disruption & Trophic Cascade Case Study: Removal of a keystone predator (e.g., Pisaster ochraceus sea star in rocky intertidal zones, or sea otters in kelp forests) triggers a top-down trophic cascade: prey populations (mussels or sea urchins) explode unchecked, competitively excluding other invertebrates or overgrazing kelp producers, collapsing Simpson's Diversity Index. Similarly, agricultural N/P fertilizer runoff causes Cultural Eutrophication: algal blooms block light, aerobic bacteria decompose dead algae and deplete dissolved O_2, creating hypoxic dead zones.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 8.7 to prove why the claim holds.

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